

Linear Convergence for General Continuous Costs in Quadratically Regularized Optimal Transport

Candidate solution packet for arXiv:2509.08547

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Status. Candidate full solution, likely valid pending expert review. The result proves the general-continuous-cost extension proposed in Remark 2.5 of the source paper, under its original weak marginal hypotheses.

1 Source question

González-Sanz, Nutz, and Riveros Valdevenito prove eventual linear L^2 convergence of dual gradient descent for quadratically regularized optimal transport with squared Euclidean cost [1]. In Remark 2.5 (page 7), they explain that the special cost is used only to show that certain zero-level sections are negligible and ask whether their theorem extends to general continuous costs. The theorem and remark are reproduced below.

Assumption 2.2. The initial values $f_0 : \Omega \rightarrow \mathbb{R}$ and $g_0 : \Omega' \rightarrow \mathbb{R}$ are Lipschitz continuous functions normalized such that $\int f_0 dP = \int g_0 dQ$. The step size η satisfies $\eta \in (0, \varepsilon)$.

We can now state the main result.

Theorem 2.3 (Linear convergence of gradient descent). *Under Assumption 2.2, there exist constants $\delta_* \in (0, 1)$ and $n_0 \in \mathbb{N}$ such that the iterates (f_n, g_n) of (16) satisfy*

$$\|(f_n, g_n) - (f_*, g_*)\|_{L^2_{\oplus}} \leq \delta_*^n \quad \text{for all } n \geq n_0.$$

The proof, which occupies the rest of the paper, has the following structure. We observe that the iterates (f_n, g_n) satisfy

$$\begin{pmatrix} f_{n+1} - f_* \\ g_{n+1} - g_* \end{pmatrix} = \mathbb{L}_n \begin{pmatrix} f_n - f_* \\ g_n - g_* \end{pmatrix}$$

for an operator $\mathbb{L}_n$ (see Lemma 5.1) which converges in operator norm to a limit $\mathbb{L}$ as $n \rightarrow \infty$ (Proposition 5.6). The limiting operator $\mathbb{L}$ is the linearization of the gradient descent operator at the optimum (f_*, g_*) . The key step is to show that the operator norm of $\mathbb{L}$ is strictly smaller than one (Proposition 4.1). It then follows that $\mathbb{L}_n$ is a strict contraction for $n \geq n_0$, which is the assertion of Theorem 2.3.

Remark 2.4. Assumption 2.2 allows for step sizes $\eta < \varepsilon$. A classical result states that for a strongly convex function with L -Lipschitz gradient, the gradient descent algorithm converges linearly for any step size $\eta < 2/L$ (see [26, Theorem 2.1.15]). In our problem, the gradient DG is Lipschitz in $L^2_{\oplus}$ with Lipschitz constant bounded by $2/\varepsilon$ (cf. Lemma 5.4). In that sense, Assumption 2.2 is in line with the classical result. Our experiments in Section 6 suggest that convergence can break down for $\eta \geq \varepsilon$.

Remark 2.5. We focus on the transport cost $c(x, y) = \frac{1}{2}\|x - y\|^2$ which is the most important example in practice. While the continuity of this function is used throughout, the specific form is used only to infer the regularity properties in Lemma 3.2 (i) which, in turn, are used to argue that certain sets occurring in the proof of Proposition 5.6 are negligible. While we do not know how to rigorously guarantee the latter in general, it seems plausible that Theorem 2.3 could extend to more general transport costs. The analysis of the linearized gradient descent operator in Section 4 directly extends to general continuous costs $c(x, y)$.

The result below answers that question affirmatively. The new argument does not prove convergence of the secant operators to a single linearization; instead, it finds a common spectral gap for all eventual secant operators.

2 Setup

Let P, Q be probability measures with compact topological supports Ω, Ω' . Assume that at least one support is connected; by symmetry we take Ω' connected. Let $c : \Omega \times \Omega' \rightarrow \mathbb{R}$ be continuous and let $\varepsilon > 0$. The QOT dual objective is

$$\Gamma(f, g) = \int f dP + \int g dQ - \frac{1}{2\varepsilon} \int (f(x) + g(y) - c(x, y))_+^2 dP(x) dQ(y).$$

We use the normalized Hilbert space

$$\mathcal{H}_0 = \left\{ (f, g) \in L^2(P) \oplus L^2(Q) : \int f dP = \int g dQ \right\}.$$

For continuous initial data $(f_0, g_0) \in \mathcal{H}_0$ and step size $0 < \eta < \varepsilon$, define

$$(f_{n+1}, g_{n+1}) = (f_n, g_n) + \eta D\Gamma(f_n, g_n).$$

The gradient belongs to $\mathcal{H}_0$, so normalization is preserved. Let (f_*, g_*) be the normalized optimal potentials and set

$$\xi_n(x, y) = f_n(x) + g_n(y) - c(x, y), \quad \xi_*(x, y) = f_*(x) + g_*(y) - c(x, y).$$

The first-order equations are

$$\int (\xi_*(x, y))_+ dQ(y) = \varepsilon \quad (x \in \Omega), \quad \int (\xi_*(x, y))_+ dP(x) = \varepsilon \quad (y \in \Omega'). \quad (1)$$

3 Proof intuition

The derivative of the positive part between ξ_n and ξ_* is an exact secant weight $w_n(x, y) \in [0, 1]$. A large zero-level set can make these weights oscillate and prevents the operator-norm limit used in the source. But no limit is needed.

Equation (1) forces a fixed strictly active core $E = \{\xi_* > \varepsilon/2\}$ to have uniformly positive sections. On the normalized space, the corresponding energy

$$\int_E (f(x) + g(y))^2 dP(x) dQ(y)$$

has a positive spectral gap. Connectedness is exactly what eliminates its kernel: a zero-energy pair is locally a pair of opposite constants along the open bipartite core, hence globally so, and normalization removes the constant.

After burn-in, both ξ_n and ξ_* are positive on E , so $w_n = 1$ there. Every exact secant energy therefore dominates the same gapped core. Its upper spectral bound is universally 2, and $0 < \eta/\varepsilon < 1$ turns the exact error recursion into a uniform strict contraction.

4 Main theorem

Theorem 1 (General continuous costs). *Let P, Q have compact supports and suppose that at least one support is connected. Let c be continuous, $(f_0, g_0) \in \mathcal{H}_0$ be continuous, and $0 < \eta < \varepsilon$. Then there exist $n_0 \in \mathbb{N}$, $C < \infty$, and $\delta \in (0, 1)$ such that*

$$\|(f_n, g_n) - (f_*, g_*)\|_{L^2(P) \oplus L^2(Q)} \leq C\delta^n \quad (n \geq n_0).$$

Thus Theorem 2.3 of [1] extends to every continuous transport cost. In particular it holds under all the marginal assumptions of that theorem; the condition that P, Q avoid boundaries of convex sets is unnecessary for this extension.

5 Proof

5.1 Uniform convergence before linearization

Lemma 2. *In the setting of Theorem 1,*

$$\|\xi_n - \xi_*\|_{L^\infty(\Omega \times \Omega')} \longrightarrow 0.$$

Proof. The source paper proves this before its only use of the squared Euclidean form; we record why the argument needs only continuity of c .

First, the elementary positive-part inequalities and $\eta/\varepsilon \leq 1$ show that the sup-norm distance of the iterates from the potentials is uniformly bounded (up to the harmless potential translation). This is the proof of Lemma 5.2 in [1] and is independent of c .

Second, the iterates have a common modulus of continuity. Indeed, for two points x, x' , write the secant identity

$$(u)_+ - (v)_+ = \theta(u - v), \quad 0 \leq \theta \leq 1,$$

inside the Q -integral defining the update of f_n . If ρ_n is a modulus for f_n and

$$\omega_c(r) = \sup_{\substack{|x-x'| \leq r \\ y \in \Omega'}} |c(x, y) - c(x', y)|,$$

then the difference of the two updates is a convex combination, with total weight at most η/ε , of a term bounded by $\rho_n(r)$ and one bounded by $\omega_c(r)$. Hence f_{n+1} has modulus at most $\max(\rho_n, \omega_c)$. The same holds for g_n . Starting from continuous data gives a common modulus for all iterates.

The map $D\Gamma$ is $2/\varepsilon$ -Lipschitz on the Hilbert space, again by the one-Lipschitz property of the positive part. Since $0 < \eta < \varepsilon$, the usual ascent inequality gives

$$\sum_{n=0}^{\infty} \|D\Gamma(f_n, g_n)\|_2^2 < \infty.$$

Uniform boundedness and equicontinuity give compactness by Arzelà–Ascoli. Every cluster point is stationary. Existence and uniqueness of continuous potentials modulo translation for a continuous QOT cost, with uniqueness under connectedness of one support, are the general-cost preliminary results used in [1]; their proof uses (1) and continuity only. Normalization therefore makes (f_*, g_*) the unique cluster point. Thus $f_n \oplus g_n \rightarrow f_* \oplus g_*$ uniformly, which is precisely the asserted convergence of ξ_n . $\square$

5.2 A fixed active core has a spectral gap

Let $B = \|(\xi_*)_+\|_\infty$. Equation (1) implies $B \geq \varepsilon$. Set

$$\tau = \frac{\varepsilon}{2}, \quad E = \{(x, y) : \xi_*(x, y) > \tau\}, \quad \lambda = \frac{\varepsilon}{2B} > 0.$$

For all $x \in \Omega$ and $y \in \Omega'$, respectively,

$$Q(E_x) \geq \lambda, \quad P(E^y) \geq \lambda. \quad (2)$$

Indeed,

$$\varepsilon = \int (\xi_*)_+ dQ \leq \tau + BQ\{\xi_* > \tau\},$$

and the other inequality is symmetric.

For $(f, g) \in \mathcal{H}_0$, define the core form

$$\mathfrak{a}_E(f, g) = \int_E (f(x) + g(y))^2 dP(x) dQ(y).$$

Lemma 3 (Active-core gap). *There is a constant $\kappa > 0$ such that*

$$\mathfrak{a}_E(f, g) \geq \kappa(\|f\|_{L^2(P)}^2 + \|g\|_{L^2(Q)}^2) \quad ((f, g) \in \mathcal{H}_0).$$

Proof. Let $q(x) = Q(E_x)$ and $p(y) = P(E^y)$. The self-adjoint operator represented by $\mathfrak{a}_E$ has diagonal multiplication part $D = \text{diag}(q, p)$ and cross operator

$$Tg(x) = \int \mathbf{1}_E(x, y)g(y) dQ(y), \quad T^*f(y) = \int \mathbf{1}_E(x, y)f(x) dP(x).$$

The kernel $\mathbf{1}_E$ belongs to $L^2(P \otimes Q)$, so T and T^* are Hilbert–Schmidt. By (2), the compression of D to $\mathcal{H}_0$ is bounded below by λ . The compressed core operator is therefore an invertible positive operator plus a compact operator, hence is Fredholm.

It remains to show that its kernel is zero. If $\mathfrak{a}_E(f, g) = 0$, then $f(x) + g(y) = 0$ for $P \otimes Q$ -almost every $(x, y) \in E$. Fix $y_0 \in \Omega'$. By (2), choose x_0 with $(x_0, y_0) \in E$. Since E is relatively open, there are relative-open neighborhoods $U \ni x_0$ and $V \ni y_0$ such that $U \times V \subset E$. Both have positive measure because Ω, Ω' are topological supports. Fubini's theorem now shows that g is Q -almost everywhere constant on V .

These neighborhoods cover the connected set Ω' . Constants agree on every nonempty overlap because such an overlap is relative-open and has positive Q -measure. Connectedness then yields $g = c$ Q -almost everywhere on Ω' . Since every E_x has positive Q -measure, the zero-energy identity gives $f = -c$ P -almost everywhere. The normalization $\int f dP = \int g dQ$ forces $c = 0$.

Thus the compressed Fredholm operator has trivial kernel. Being self-adjoint of index zero, it is invertible; its positive quadratic form consequently has some lower bound $\kappa > 0$. $\square$

5.3 Every eventual secant operator dominates the core

Let $e_n = (f_n - f_*, g_n - g_*) \in \mathcal{H}_0$. Define the measurable secant slope

$$w_n(x, y) = \begin{cases} \frac{(\xi_n(x, y))_+ - (\xi_*(x, y))_+}{\xi_n(x, y) - \xi_*(x, y)}, & \xi_n(x, y) \neq \xi_*(x, y), \\ \mathbf{1}_{\{\xi_*(x, y) > 0\}}, & \xi_n(x, y) = \xi_*(x, y). \end{cases}$$

Then $0 \leq w_n \leq 1$, and the gradient update gives the exact recurrence

$$e_{n+1} = \left(I - \frac{\eta}{\varepsilon} A_n\right) e_n, \quad (3)$$

where A_n is the self-adjoint operator on $\mathcal{H}_0$ with form

$$\langle A_n(f, g), (f, g) \rangle = \int w_n(x, y) (f(x) + g(y))^2 dP(x) dQ(y).$$

By Lemma 2, for all sufficiently large n we have $\|\xi_n - \xi_*\|_\infty < \tau$. On E , both $\xi_* > \tau$ and $\xi_n > 0$, so $w_n = 1$. Lemma 3 then gives

$$\langle A_n h, h \rangle \geq \kappa \|h\|^2 \quad (h \in \mathcal{H}_0). \quad (4)$$

For the upper bound, write $m = \int f dP = \int g dQ$. Since $w_n \leq 1$,

$$\begin{aligned} \langle A_n(f, g), (f, g) \rangle &\leq \int (f(x) + g(y))^2 dP(x) dQ(y) \\ &= \|f\|_2^2 + \|g\|_2^2 + 2m^2 \\ &\leq 2(\|f\|_2^2 + \|g\|_2^2). \end{aligned}$$

Thus the spectrum of every eventual A_n is contained in $[\kappa, 2]$. Putting $a = \eta/\varepsilon \in (0, 1)$, the spectral theorem and (3) yield

$$\|e_{n+1}\| \leq \delta \|e_n\|, \quad \delta = \max\{|1 - a\kappa|, |1 - 2a|\} < 1.$$

Iteration proves the stated geometric estimate. $\square$

6 Relation to later literature

The later PL-inequality paper [2] independently proves linear convergence for arbitrary uniformly continuous costs and provides explicit rates, but assumes that one marginal has a density bounded above and away from zero on a convex support (or on the closure of a bounded Lipschitz domain). Theorem 1 is non-quantitative but retains the source paper’s substantially weaker marginal framework; its active-core Fredholm argument requires only compact supports and connectedness of one support.

7 Verification, limitations, and novelty

The proof was audited step by step in `VERIFICATION.md`. The optional matrix checker `code/verify_weighted_core.py` verifies the spectral domination and contraction inequalities for finite weighted bipartite cores; it is not part of the proof.

The theorem gives a qualitative contraction factor because the Fredholm gap κ is not explicit. It concerns fixed-step dual gradient descent with $0 < \eta < \varepsilon$ and does not assert convergence at or above the threshold.

A bounded literature search on 9 August 2026 used the exact source title and arXiv id together with “general continuous cost,” “strictly active,” “secant weights,” “spectral gap,” and “linear convergence.” It found the source and the stronger-assumption PL result [2], but no theorem with the weak marginal hypotheses above and no use of the present secant-domination argument. Novelty confidence is moderate pending specialist review.

8 Human-review recommendation

The packet is ready for expert review as a candidate full solution of Remark 2.5. The principal checks are the local-a.e.-constancy patching in Lemma 3, the Fredholm compression to $\mathcal{H}_0$, and the claim that the source’s preconvergence proof remains valid for every continuous cost.

References

- [1] A. González-Sanz, M. Nutz, and A. Riveros Valdevenito, *Linear convergence of gradient descent for quadratically regularized optimal transport*, arXiv:2509.08547v3, 2026.
- [2] A. González-Sanz, M. Nutz, and A. Riveros Valdevenito, *Polyak–Łojasiewicz inequality for quadratically regularized optimal transport*, arXiv:2605.27175, 2026.